

Problem 18.2

Which of the following are equal to $a_{\overline{3}|}$?

(i) $v + v^2 + v^3$

(ii) $\ddot{a}_{\overline{4}|} - 1$

(iii) $a_{\overline{\infty}|}(1 - v^3)$

Solution

Recall that

$$a_{\overline{3}|} = v + v^2 + v^3 = \frac{1 - v^3}{i}.$$

We check each expression.

Statement (i)

By definition,

$$a_{\overline{3}|} = v + v^2 + v^3.$$

Therefore, statement (i) is equal to $a_{\overline{3}|}$.

Statement (ii)

Since

$$\ddot{a}_{\overline{4}|} = 1 + v + v^2 + v^3,$$

we have

$$\ddot{a}_{\overline{4}|} - 1 = v + v^2 + v^3 = a_{\overline{3}|}.$$

Therefore, statement (ii) is equal to $a_{\overline{3}|}$.

Statement (iii)

The present value of a perpetuity-immediate of 1 is

$$a_{\overline{\infty}|} = \frac{1}{i}.$$

Thus,

$$a_{\overline{\infty}}(1 - v^3) = \frac{1}{i}(1 - v^3) = \frac{1 - v^3}{i} = a_{\overline{3}}.$$

Therefore, statement (iii) is also equal to $a_{\overline{3}}$.

Hence, all three statements are equal to $a_{\overline{3}}$:

$$\boxed{\text{(i), (ii), and (iii)}}.$$